Dissertation Project Brief: GDP vs Night Lights in Sub-Saharan Africa

Handover document for a new chat. State as of 28 October 2026. Student: Ziyan Rafiq (zr388), BSc International Development with Economics, University of Bath. Unit SP32038 (10,000 words; deadline 22 April 2027). Supervisor: Ricky Kanabar (rk735).

One-line summary. A balanced panel of 40 Sub-Saharan African countries (2016–2021, 240 country-years) tests whether the gap between officially reported GDP growth and satellite night-lights growth is larger where statistical capacity (World Bank SPI, rebuilt) and political accountability (V-Dem) are lower, using country and year fixed effects; Ghana and Kenya are case studies.

Status legend used throughout: **Confirmed** = checked against the actual data or document in the previous chat; **Caveat** = checked but with a known issue; **Open** = not yet done or not verified. Literature citations were given from memory and *must all be verified* before use.

1 Research question, aims, scope

Research question (agreed with supervisor): *What explains divergence between officially reported GDP growth and satellite-observed night-time economic activity in Ghana and Kenya? Evidence from a Sub-Saharan African panel.*

- Ricky's original suggestion was the Ghana–Kenya question, framed so findings have implications for developing countries. He agreed that 12 observations (2 countries × 6 years) is far too few: "you either need more granular data ... or more countries. In your case, it's going to be more countries."

Aims

1. Measure the gap between official GDP growth and night-lights growth across 40 SSA countries, 2016–2021.
2. Test whether the gap is larger where statistical capacity (SPI) or political accountability (V-Dem) is lower.
3. Explain Ghana's and Kenya's year-by-year gaps using annual data on GDP rebasing dates, electricity access and oil output. Why these two: they contrast on capacity and accountability, and each faces a main measurement problem (Kenya: fastest electrification in the region, +23 pp 2016–21; Ghana: offshore oil and a 2019 rebasing). Ricky asked what aim 3 means; answer: not comparing SPI/V-Dem to events, but explaining each country's residual/gap with annual event data.

Rationale: GDP guides aid, debt and policy; African national accounts are often unreliable (Jerven); satellites give an independent check. **Scope caveats:** divergence ≠ deliberate manipulation; lights are an imperfect proxy (miss agriculture/informal/off-grid; pick up electrification, gas flares). Ricky wants an explicit justification of why luminosity is a sufficiently good measure of income and development, and a discussion of problems with GDP as a measure.

2 Data and sample

2.1 Sample construction (Confirmed)

Step	Countries	Rows
All Sub-Saharan Africa (World Bank region definition, from SPI file)	48	288
Unbalanced: drop individual missing country-years	43	252
Balanced (main, supervisor's recommendation): drop any country with any missing value	**40**	**240**

Balanced = 288 − (8 countries × 6) = 240. (A common mistake is 252 − 48 = 204: wrong, because 36 of those 48 rows were already excluded from 252; dropping the 8 countries fully loses only 12 extra rows: DR Congo 5, Seychelles 5, Equatorial Guinea 2.)

Dropped country	Years missing (of 6)	Missing variable(s)
Eritrea	6	GDP growth, agriculture share, SPI
South Sudan	6	GDP growth, agriculture share
Somalia	6	Agriculture share (SPI also missing 2016)
Comoros	6	SPI
Central African Republic	6	SPI (2016–2021)
Equatorial Guinea	4	SPI (2016–2019)
DR Congo	1	SPI (2016)
Seychelles	1	SPI (2016)

Exact missing counts, 2016–2021 (288 possible rows): GDP growth 12 (ERI, SSD all years); agriculture share 18 (ERI, SOM, SSD all years); SPI 25 (CAF, COM, ERI all; GNQ 2016–19; COD, SOM, SYC 2016); V-Dem, electricity, urbanisation, population 0.

40 countries kept: Angola, Benin, Botswana, Burkina Faso, Burundi, Cabo Verde, Cameroon, Chad, Congo Rep., Côte d'Ivoire, Eswatini, Ethiopia, Gabon, Gambia, Ghana, Guinea, Guinea-Bissau, Kenya, Lesotho, Liberia, Madagascar, Malawi, Mali, Mauritania, Mauritius, Mozambique, Namibia, Niger, Nigeria, Rwanda, São Tomé and Príncipe, Senegal, Sierra Leone, South Africa, Sudan, Tanzania, Togo, Uganda, Zambia, Zimbabwe.

Survivorship bias (Ricky: must be made transparent if there is evidence): most dropped countries are fragile/conflict-affected with weak statistics, so the sample leans to stronger capacity, likely understating any capacity effect; results may not generalise to fragile states. Unbalanced panel (252 rows, 43 countries) = robustness check. Ricky's slides: insisting on a balanced panel can itself introduce survivorship bias; FE handles unbalanced panels.

Period: main 2016–2021 (SPI overall score starts 2016; VIIRS v2.1 covers 2012–2021). 2022–2023 only as a robustness check (VIIRS version switch; oil rents end 2021). Ricky: dates "fine"; short panel helps with the slow-changing-FE issue (Millimet & Bellemare 2025). Growth for 2016 needs 2015 levels (lights and GDP available).

File produced: `ssa_balanced_panel_2016_2021.csv` (240 rows). Columns: iso3c, country, year, gdp_growth, gdp_real_lcu, agri_share, urban_share, population, elec_access, elec_access_change (uses 2015), spi_overall, spi_p1_use, spi_p2_services, spi_p3_products, spi_p4_sources, spi_p5_infra, v2x_accountability, v2x_veracc, v2x_diagacc, v2x_horacc. **Not yet in it:** lights growth, rebuilt SPI, COVID stringency, oil rents, rebasing dummy.

2.2 Data sources and verification results

Variable	Source / code	Verification and notes
GDP growth	WDI `NY.GDP.MKTP.KD.ZG`; levels `NY.GDP.MKTP.KN` (constant LCU)	**Confirmed** Last updated 2026-10-08 (record vintage). KD.ZG equals growth computed from KN exactly (max diff 0.00). No visible rebasing break in Ghana/Kenya growth (series appears back-revised; inference, check WDI notes). Prefer computing log-differences from KN.
Night lights	EOG VIIRS annual VNL, v2.1, masked average radiance	**Open** Not yet processed: student computes it (download annual rasters 2015–2021, sum radiance within national borders, take log change). Nothing in the provided files gives lights growth. See Section 2.3.
Statistical capacity	World Bank SPI. Index file `03_output_data/SPI_index.csv` from `https://github.com/worldbank/SPI`; indicator-level `SPI_data.csv` (uploaded)	**Confirmed** Overall score 2016–2024; SSA countries with score: 41 (2016), 44 (2017–19), 45 (2020–21), 46 (2022–23). Overall = simple mean of 5 pillars (checked exactly). Must be rebuilt (Section 3).
Accountability	V-Dem v16 (published March 2026), Country-Year Full+Others CSV (406 MB); `v2x_accountability`	**Confirmed** Covers 1789–2025; complete for SSA 2016–2023. Sub-components: `v2x_veracc` (vertical), `v2x_horacc` (horizontal), `v2x_diagacc` (diagonal); `_osp` versions roughly 0–1; uncertainty bounds in `_codelow`/`_codehigh`. V-Dem's own region code includes Zanzibar (ZZB) and Somaliland (SML), exclude.
Agriculture share	WDI `NV.AGR.TOTL.ZS`	**Confirmed** Gaps only ERI, SOM, SSD.

Variable	Source / code	Verification and notes
Electricity access	WDI `EG.ELC.ACCS.ZS` (ESMAP / SDG7 Global Electrification Database)	**Confirmed** Complete for all 40, 2015–2024. Survey-based with modelled values between surveys (limitation). Use the yearly change (pp) as control.
Urbanisation	WDI `SP.URB.TOTL.IN.ZS`	**Confirmed** Complete. Slow-moving; little within variation under FE.
Population	WDI `SP.POP.TOTL` (log)	**Confirmed** Complete.
COVID stringency	Oxford COVID-19 Government Response Tracker, `data/OxCGRT_nat_latest.csv` from `https://github.com/OxCGRT/covid-policy-tracker`, `StringencyIndex_Average` (0–100)	**Caveat** Daily 2020-01-01 to 2023-02-28; full days for 2020 and 2021. **38 of 40**: missing Guinea-Bissau and São Tomé and Príncipe. Annual means: Ghana 42.5 (2020), 42.8 (2021); Kenya 58.0, 56.1; SSA mean 46.6, 44.0. Recommendation: use only in a robustness check (keeps 40 countries); set 0 for 2016–2019.
Oil rents	WDI `NY.GDP.PETR.RT.ZS` (Changing Wealth of Nations)	**Caveat** Complete for all 40, 2016–2021; series ends 2021 (no 2022–23). 19 countries zero every year. Producers (% GDP, 2016–21): Congo Rep. 13–38, Angola 19–31, Gabon 9–18, Chad 7–17, Nigeria 3–8, Ghana 1.0–4.2, Sudan 0.6–3.4, Cameroon 1.3–2.8. Measures value (price × output − cost), so moves mainly with oil prices (dips 2016, 2020). For the Ghana case study use GSS oil/non-oil GDP instead.
Rebasing dates	SPI raw column `RAW.D5.2.2.NABY` (base year by country-year) + statistics-office reports	**Caveat** Among the 40 countries: 23 base-year changes in 2016–21 (28 events in 24 countries in 2016–22); across all 48 SSA countries 24 countries rebased 2016–21. Fixes needed: ETH, UGA, GIN, SDN store text ("2015/2016", "2010/11", "chained") to convert by hand; Liberia goes backwards 2000→1992 in 2019 (probable data error); Kenya's 2016 base year appears in SPI only from 2022, but KNBS released the revision Sept 2021, so use official release dates. Ghana: 2006→2013 in 2019 (also SNA 1993→2008).

2.3 VIIRS night lights: versions and processing (Confirmed from EOG v2.2 readme, dated 2023-07-07)

- v2.1 = annual composites 2012–2021. v2.1 fixed an annual-averaging bug in v2 and removed v2's low-detection filter (cells detected once/twice with radiance <0.7/0.5 nW). Never mix with v2.
- v2.2 = 2022 composite: mixes satellites (11 months SNPP, excluding Aug 2022; adds NOAA-20 for May, Aug, Sep 2022); includes months with >1 cloud-free coverage (v2/v2.1 required ≥3); excludes grid cells lacking median values. Implication: a 2021→2022 jump could be artificial and differ by country, so main analysis uses 2016–2021 (v2.1 only). The readme predates 2023/2024 files; their version is unverified.
- Masking steps: background data-range threshold, aurora filtering, lit-cell mask. **No gas-flare removal mentioned**, so flares must be handled (offshore flares fall outside land borders if land boundaries are used; check Ghana's offshore fields fall outside the polygon). Method paper: Elvidge et al. (2021), *Remote Sensing* 13(5): 922.
- Access: EOG reportedly requires a free account for v2.2 downloads (unofficial source; register early). Google Earth Engine also hosts `NOAA/VIIRS/DNB/ANNUAL_V21` and `_V22`.
- Processing plan (Python): `rasterio` + `rasterstats`/`exactextract` + `geopandas` (country boundary file with ISO3 codes). Sum masked average radiance within each border per year; $\Delta \ln L_{it} = \ln L_{it} - \ln L_{i,t-1}$. Set a rule for zero/near-zero totals before seeing results; a country with missing/near-zero lights would be dropped to keep the panel balanced.

3 Statistical capacity: SPI and why it is rebuilt

What SPI is: World Bank Statistical Performance Indicators (replaced the Statistical Capacity Index, SCI, published since 2004; SCI discontinued after 2020). Score 0–100 from about 51 indicators in 5 pillars: (1) data use, (2) data services, (3) data products, (4) data sources, (5) data infrastructure. SPI measures *performance* (outputs, compliance with standards), not only *capacity*. Methodology paper: Dang, Pullinger, Serajuddin & Stacy (2023), *Scientific Data* (verify).

2025 SPI update note (uploaded PDF, 24 Nov 2025): data through 2024; 188 economies in 2024 (167 in 2016); no methodological change in this vintage; revisions mainly Pillars 1 and 4 with minimal impact (2022 correlation previous vs revised 0.98–0.99); global average +12.5 points 2016–2024; rank correlation 0.9; jump in

2021 (Pillar 3 69.8→79.1, business census indicator 0.15→0.79), so year effects are needed.

Circularity problem: Pillar 5, Dimension 5.2 "Standards and Methods" (10 indicators) includes two that describe the GDP series itself:

Indicator	Official scoring (SPI metadata, `SPI_full_metadata.csv`)
5.2.1 System of national accounts in use (`SNAU`)	1 = SNA2008/ESA2010; 0.5 = SNA1993/ESA1995; 0 otherwise
5.2.2 National accounts base year (`NABY`)	1 = chained prices; 0.5 = base year within past 10 years; 0 otherwise

SPI therefore falls mechanically as a base year ages past 10 years and jumps back at rebasing, exactly when GDP figures are revised. Ghana NABY: 0.5 (2016) → 0 (2017–18, base 2006) → 0.5 (2019+, base 2013); SNAU 0.5→1 in 2019. Kenya NABY: 0.5 (2016–19) → 0 (2020–21) → 0.5 (2022+). Pillar 5: Ghana 45→60, Kenya 45→70. Rough magnitude: one 0.5 step moves overall SPI by about 0.3–0.7 points (vs within-country SD 5.5): small but systematic. Ricky agreed: "SPI should just measure capacity given what you need it for." Other 5.2 indicators kept (CPI base year `CPIBY` is borderline: drop in one robustness check). The SPI metadata itself admits validity concerns about this dimension.

Rebuild method: follow the SPI team's R code `02_programs/02-SPI_index.Rmd` in the GitHub repo: Dimension 5.2 = mean of its indicators (line 1398); Pillar 5 = *weighted* average of its dimensions (lines 1434–1442); overall = mean of 5 pillars. First reproduce the published `SPI.INDEX` exactly, then drop `SPI.D5.2.1.SNAU` and `SPI.D5.2.2.NABY`. **Open** A first naive reconstruction (simple means) did *not* match the published scores, so use the dimension weights.

Capacity measures: (1) Rebuilt SPI = main; (2) SPI Pillar 4 (data sources) = alternative "data-collection capacity"; (3) original SPI = robustness. Pillar 4 contents: 4.1 censuses (population & housing, agriculture, business; 1 if within 10 years, 0.5 within 20) and surveys (household income/expenditure, agriculture, labour force, health/DHS, business; 1 = 3+ in 10 years, 0.6 = 2, 0.3 = 1); 4.2 admin data (in the index only CRVS: births ≥90% and deaths ≥75% registered); 4.3 geospatial (ODIN sub-national availability). Pillar 4 caveats: step changes when censuses age past 10 years; possible under-counting (scores depend on international microdata catalogues; Kenya scores 0 on labour-force and agriculture surveys, which should be checked against KNBS). Pillar 4 totals: Ghana 45.3 (2016), 54.5 (2019), 57.2 (2023); Kenya 35.9, 45.8, 46.5; rise mostly from new agriculture censuses (Ghana 2018, Kenya 2019). Correlation overall SPI vs Pillar 4, SSA 2016–21: 0.78.

Variation (SSA, 2016–23): overall SPI within-country SD 5.5 vs between 10.9; Pillar 1 9.1/20.2; P2 12.1/13.2; P3 7.2/8.6; P4 4.2/14.2; P5 10.9/13.4. SPI change 2016–21: median 13.1 points; Kenya 13.1, Ghana 6.8; largest Zimbabwe 31.1, Senegal 22.4, Burkina Faso 21.0, Gambia 20.8, Liberia 20.6 (Somalia 29.8 but dropped). Zimbabwe's 2019 currency changes may distort its GDP: check as an outlier.

4 Political accountability: V-Dem

V-Dem (University of Gothenburg): expert surveys from 4,200+ country experts combined with a Bayesian measurement model; 500+ indicators, 202 countries. `v2x_accountability` answers: to what extent are those in government held accountable? It combines vertical (elections, parties), horizontal (courts, legislature, oversight bodies) and diagonal (media, civil society, free expression) accountability. Scale roughly −2 to +2 (Ghana about 1.2); standardise or use `_osp` (0–1) and justify. Diagonal accountability is closest to "who would catch misreporting" (Martinez's mechanism).

Variation (SSA 2016–23), within vs between SD: overall 0.141 vs 0.667; vertical 0.196 vs 0.575; diagonal 0.155 vs 0.697; horizontal 0.201 vs 0.792. Between variation is 3–5× larger, so under FE the accountability effect is identified from a few large movers. Changes 2016–21: rises Gambia +1.31 (democratic transition), Ethiopia +0.45, Burundi +0.29, Sierra Leone +0.25; falls Benin −0.85, Mali −0.52 (coups), Mauritius −0.46, Côte d'Ivoire −0.32 (Comoros, Seychelles also large movers but dropped). These events may coincide with economic shocks: discuss; robustness without the biggest movers.

5 Regression specification (current, after Ricky's 28 Oct feedback)

Outcome = the gap. Ricky asked why the outcome was not the gap, noting the units match. Both are log changes (≈ % growth), so they can be subtracted:

$$Gap_{it} = \Delta \ln GDP_{it} - \Delta \ln L_{it} \tag{1}$$

$Gap_{it} > 0$: reported GDP grew faster than lights.

Main specification (country and year fixed effects; SEs clustered by country, 40 clusters):

$$\begin{aligned} Gap_{it} = \alpha + \theta\, SPI_{it} + \phi\, Acc_{it} + \pi_1 Agr_{it} + \pi_2\, \Delta Elec_{it} \\ + \pi_3\, Urb_{it} + \pi_4 \ln Pop_{it} + \pi_5\, Covid_{it} + \mu_i + \lambda_t + \varepsilon_{it} \end{aligned} \tag{2}$$

Estimated twice: signed gap Gap_{it} (overstatement; expect $\phi < 0$: less accountability → GDP grows faster than lights) and absolute gap $|Gap_{it}|$ (noise; expect $\theta < 0$: more capacity → smaller gaps either way). Note: $Covid_{it}$ is proposed for the robustness check only (2 countries missing).

Check model (lights on the right-hand side):

$$\begin{aligned}\Delta \ln GDP_{it} = \alpha + \beta\, \Delta \ln L_{it} + \gamma\, (\Delta \ln L_{it} \times SPI_{it}) + \delta\, (\Delta \ln L_{it} \times Acc_{it}) \\ + \theta\, SPI_{it} + \phi\, Acc_{it} + X'_{it}\pi + \mu_i + \lambda_t + \varepsilon_{it}\end{aligned} \tag{3}$$

The gap model is the special case $\beta = 1$, $\gamma = \delta = 0$. Why it matters: the gap assumes GDP and lights move one-for-one, but the literature finds GDP moves less than one-for-one (elasticity about 0.3 in Henderson, Storeygard & Weil 2012, older DMSP data, from memory). Illustration: lights +20%, GDP +6% gives simple gap −14 although 6% is exactly what $\beta = 0.3$ predicts; lights +5%, GDP +6% gives gap +1 although GDP is 4.5 points above expected. Country FE absorb each country's average gap but not year-to-year swings. Decision rule: estimate $\hat{\beta}$, t-test H_0: $\beta = 1$; if not rejected, the simple gap is justified; if well below 1, give the check model (or a scaled gap $\Delta \ln GDP - \hat{\beta}\, \Delta \ln L$, two-step, SEs need care) more weight. No extra data is needed: coefficients are estimated from the same columns; interaction terms are products of existing columns. Centre SPI and Acc before interacting so β is the elasticity at average capacity/accountability; the link for a country-year is $\beta + \gamma SPI + \delta Acc$.

Notation: i country (1–40); t year (2016–2021); α constant; θ, ϕ effects of capacity and accountability on the gap; π_1–π_5 control coefficients ($X'\pi$ is shorthand for the sum of controls times coefficients; controls are there to avoid bias and are not the focus); β lights elasticity; γ, δ how it changes with SPI and Acc; μ_i country FE (unchanging traits and each country's average gap); λ_t year FE (common shocks incl. COVID 2020); ε_{it} error = unexplained part, which is also the "divergence" examined in the case studies. Why logs: changes in logs ≈ % growth; β becomes an elasticity comparable across country sizes; reduces skew; matches the literature; $\ln 0$ undefined (rule needed for near-zero lights). Why cluster: a country's errors are correlated across years; ignoring this understates SEs (slides' `cluster(idcode)` example roughly doubles SEs).

Previous version (before 28 Oct): outcome was $\Delta \ln GDP_{it}$ with lights on the RHS and interactions (the current check model). Changed because the gap answers the RQ directly, is easier to read (Ricky's preference), and separates overstatement from noise.

Estimators (from Ricky's AQM1 Lecture 3 slides): Pooled OLS with clustered SEs (baseline); **fixed effects (main)**: within transformation; Ricky: "yes to FE in general, your panel is relatively short which helps"; random effects + Hausman test (if FE and RE differ significantly, keep FE); first differences (FE vs FD: with serially correlated errors FD is better; Millimet & Bellemare 2025 recommend comparing FE/FD for $T > 2$). FE caveats from the slides: cannot estimate time-invariant variables; imprecise when regressors barely vary within units (V-Dem); worsens measurement error (key limitation: both GDP and lights are noisy); strict exogeneity required. Slides use one-way (individual) effects; year dummies added on top is an extension. Python: `linearmodels` (`PooledOLS`, `PanelOLS(entity_effects=True, time_effects=True)`, `RandomEffects`, `FirstDifferenceOLS`); Hausman not built in: $H = (b_{FE} - b_{RE})'[V_{FE} - V_{RE}]^{-1}(b_{FE} - b_{RE})$. Ricky only uses Stata and will not answer programming questions.

Robustness checks planned: original SPI; SPI Pillar 4; drop CPI base year; check model / test $\beta = 1$; rebasing-year dummy (Ghana 2019, Kenya 2021 via KNBS date; panel-wide from base-year changes); $\Delta \ln L \times \Delta Elec$ interaction; COVID stringency control (38 countries) and 2016–2019 only; excluding Kenya; excluding Ghana; excluding oil producers (Nigeria, Angola, Gabon, Congo Rep., Chad; also Ghana) or adding oil rents; excluding biggest accountability movers; unbalanced panel (252 rows); 2016–2023 including VIIRS v2.2; RE + Hausman; FD.

COVID-19 (Ricky asked how to model): year FE absorb the common 2020–21 shock; lockdowns differed by country, so add OxCGRT stringency (robustness) and re-estimate on 2016–2019.

6 Ghana and Kenya: facts gathered

	Ghana	Kenya
GDP growth (WDI), 2016–2023	3.37, 8.13, 6.20, 6.51, 0.51, 5.08, 3.80, 3.14	4.21, 3.84, 5.65, 5.11, −0.27, 7.59, 4.86, 5.72
SPI overall 2016 / 2021 / 2023	59.3 / 66.2 / 66.2	55.9 / 68.1 / 70.6 (overtakes Ghana from 2021)
V-Dem accountability 2016 / 2023	1.36 / 1.16	0.96 / 1.03
Vertical / diagonal / horizontal 2023	0.95 / 1.18 / 1.27	0.71 / 1.24 / 1.14
Electricity access 2015 / 2021 / 2023	74.0 / 86.3 / 89.5%	41.6 / 76.5 / 76.2% (largest rise in SSA, +23.4 pp 2016–21; SSA median +6.5)
COVID stringency 2020 / 2021	42.5 / 42.8	58.0 / 56.1
Oil rents (% GDP)	1.0 (2016) to 4.2 (2018)	≈ 0
GDP rebasing history	2010: base 1993→2006, GDP +60%; base 2013 (in SPI from 2019, plus SNA2008); next rebasing targeted mid-2027	2014: base 2001→2009, GDP +25.4%; Sept 2021: base 2009→2016, about +5.3% (secondary source, Cytonn)

Implications: Under FE, Kenya contributes much to identifying θ/γ (large SPI change) while both contribute little to ϕ/δ (accountability converges). Kenya: SPI rise, electrification surge and rebasing coincide in 2020–22, so control for $\Delta Elec$, add the interaction and run leave-one-out; Kenya would otherwise look like GDP under-reports. Ghana: offshore oil adds GDP but not land lights, so use GSS non-oil GDP for the case study; rebasing is a panel-wide issue (half the region). Case-study method: plot each country's gap and the model's prediction by year; match residuals to documented events (GSS, KNBS, IMF sources); cautious language ("consistent with").

Sources found: IMF report on Kenya's rebased GDP (2022) `https://imf.org/-/media/Files/Publications/CR/2022/English/1KENEA2022001.ashx`; KNBS press release `https://www.knbs.or.ke/press-release-on-launch-of-the-revised-national-accounts-estimates/`; KNBS quarterly GDP `https://www.knbs.or.ke/quarterly-gdp/`; GSS 2010 rebasing newsletter (statsghana.gov.gh); GSS quarterly GDP bulletins (with non-oil GDP); Business Daily on Kenya 2014 (+25.4%).

7 Literature review

Department guidance (Week 4 slides): 3–4 themes (ideally 3); 3,000–3,500 words; structure options: history, policy topic, complexity (basics then complications), popularity, own perspective (no "I think"); be selective, analytical, set parameters; conclusion 200–250 words stating gaps and what the dissertation adds; subheadings; link back to the RQ.

Themes (restructured per Ricky, 28 Oct):

1. **Measuring growth where official statistics are weak, and night lights as an alternative** (merged old themes 1 and 2): limits of GDP as a measure of income/development; why African GDP is unreliable (capacity, informality, outdated base years); lights as a proxy, why luminosity is a sufficiently good measure of income and development, when lights add most value, what they miss or wrongly pick up; DMSP vs VIIRS. Key: Jerven (2013) *Poor Numbers*; Jerven & Duncan (2012, *African Statistical Journal*); Devarajan (2013, *Review of Income and Wealth*); Sandefur & Glassman (2015, *J. Development Studies*); Young (2012, *JPE*); Henderson, Storeygard & Weil (2012, *AER*); Chen & Nordhaus (2011, *PNAS*); Pinkovskiy & Sala-i-Martin (2016, *QJE*); Donaldson & Storeygard (2016, *JEP*); Gibson, Olivia & Boe-Gibson (2020, *J. Economic Surveys*); Gibson et al. (2021, *JDE*); Keola, Andersson & Hall (2015, *World Development*); Weidmann & Schutte (2017, *J. Peace Research*: lights predict DHS household wealth); Elvidge et al. (2021).
2. **Why reported GDP and lights diverge: capacity vs incentives**: statistical capacity and its measurement (SPI, circularity); political incentives to overstate growth. Key: Dang et al. (2023); Martinez (2022, *JPE* 130(10): 2731–2769; autocracies overstate growth by about 35%; NTL elasticity of GDP larger in autocracies; 184 countries; Freedom House regime measure: abstract verified, specification not yet read); Magee & Doces (2015, *ISQ*); Hollyer, Rosendorff & Vreeland (transparency); Wallace (2016, *BJPolS*).
3. **Alternative data for development**: satellites, mobile-phone and other non-traditional data; nowcasting. Key: Donaldson & Storeygard (2016); Jean et al. (2016, *Science*).

GDP rebasing and data revisions: moved to the **methods** chapter (Ricky). **Gap/contribution:** Martinez links divergence to regime type, not statistical capacity; Jerven's critique is mostly qualitative; most lights studies use DMSP, this uses VIIRS 2016–21; corrects SPI circularity. **Parameters:** SSA only; VIIRS era; national level; peer-reviewed plus grey literature (World Bank, IMF, NSOs). Methods references: Wooldridge (panel econometrics); Millimet & Bellemare (2025); Hale et al. (2021, *Nature Human Behaviour*) for OxCGRT.

8 Supervisor feedback log

Round 1 (around 8–20 Oct): VIIRS versions fine; be precise about gaps ("one could simply drop those countries"); survivorship should be transparent; 252 rows confirmed as rows, lights add columns not rows; 12 obs insufficient; implications of Ghana/Kenya differences; yes to FE; why SPI includes base year (agreed: rebuild); SSA panel with focal cases "definitely, yes"; "tech up", slides suffice; no programming help. Later: use a **balanced** panel. Lecturer also asked for a plan document (rows, themes, method).

Round 2 (28 Oct, on the plan PDF): justify luminosity as a measure of income/development; GDP measurement issues (rationale fine); clarify aim 3 and data for "other" factors; how to model COVID; "night lights growth: do you not simply compute this yourself?"; outcome should be the gap (same units; could rearrange); themes: merge 2 into 1, old 3 → 2, theme 4 (rebasing) → methods, theme 5 → 3; wants regression spec in Greek (LaTeX or Word). "Good progress." Reply drafted and sent with `Regression_Specification.pdf` and `Dissertation_Plan_GDP_vs_Night_Lights_v2.pdf`.

9 Decisions log

Decision	Reason
SSA panel, Ghana/Kenya as cases	12 observations too few; supervisor agreed
Balanced panel, 40 countries, 240 rows	Supervisor recommendation; unbalanced as robustness
Main period 2016–2021	SPI starts 2016; VIIRS v2.1 ends 2021; oil rents end 2021
Outcome = gap (signed and absolute)	Supervisor; matches RQ; separates overstatement from noise
FE main; RE/Hausman, FD, pooled OLS as checks	Slides; short panel; supervisor "yes to FE"
Rebuilt SPI (no SNAU, NABY)	Circularity with GDP rebasing
Python for the dissertation; R in a later module	Student's existing Python; R learnt in another module (could replicate with `fixest` next semester)
COVID stringency only as a robustness check (proposed)	Missing GNB, STP

10 Open tasks

1. Download VIIRS v2.1 annual composites 2015–2021 (and v2.2 for 2022–23), get a boundary file, compute lights growth; handle flares and near-zero totals; confirm the panel stays at 40.
2. Rebuild SPI with the official weights; validate against `SPI.INDEX`; also build Pillar 4 and no-CPIBY variants.
3. Build the rebasing dummy: convert text base years (ETH, UGA, GIN, SDN), check Liberia, use KNBS Sept 2021 for Kenya; check WDI growth for breaks in rebasing years (compare Kenya WDI vs KNBS revised series).
4. Add OxCGRT annual stringency (0 for 2016–19) and oil rents; get GSS oil/non-oil GDP for Ghana.
5. Decide accountability scale (standardised vs `_osp`); centre SPI and Acc.
6. Read Martinez (2022) for his exact specification (SSRN 3093296); verify every citation; start the literature review.
7. Ethics form (secondary, public, country-level data; low risk); Week 6 synopsis presentation (topic, methodology, aims, rationale, ethics).
8. Learn panel methods from Ricky's Lecture 3 (within/between variation, pooled OLS, FE, RE, Hausman); the student's background covers descriptive statistics, inference, hypothesis testing, chi-square, correlation and linear regression.

11 Files and working preferences

Files from the previous chat: WDI CSVs (GDP growth, GDP constant LCU, agriculture, urban, population, electricity, oil rents; all updated 2026-10-08); `V-Dem-CY-Full_Others-v16.csv`; `SPI_data.csv` (indicator level, 2004–2024); SPI 2025 update note PDF; VIIRS v2.2 readme (pptx); Ricky's AQM1 Lecture 3 slides (panel data); Week 4 dissertation slides (literature review); `ssa_balanced_panel_2016_2021.csv`; `Regression_Specification.pdf`; `Dissertation_Plan_GDP_vs_Night_Lights_v2.pdf` (plan doc also exists as an editable Claude doc). `SPI_index.csv` and OxCGRT data come from GitHub (links above).

How the student likes to work: concise, bullet-point answers; plain explanations of econometrics; for code, hints and pointers to concepts/tools rather than full solutions unless asked, and help debugging; wants to understand every line and to be told when something looks like unexamined copy-paste; clearly separate what is verified from what is assumed or recalled from memory; short emails to the supervisor.